

Permutations

Date _____ Period _____

List all possible permutations.

1) 1, 2, 3

123 213 312
132 231 321

2) T, V, W, taken two at a time

TV VT WT
TW VW WV

3) ☺, ☀, ♥, ▲, taken two at a time

☺☀ ☀☺ ♥☺ ▲☺
☺♥ ☀♥ ♥☀ ▲☀
☺▲ ☀▲ ♥▲ ▲♥

4) A, B, C, D

ABCD	BACD	CABD	DABC
ABDC	BADC	CADB	DACB
ACBD	BCAD	CBAD	DBAC
ACDB	BCDA	CBDA	DBCA
ADBC	BDAC	CDAB	DCAB
ADCB	BDCA	CDBA	DCBA

Evaluate each expression.5) ${}_4P_2$

12

6) ${}_5P_1$

5

7) ${}_6P_2$

30

8) ${}_6P_6$

720

9) $-4 + {}_7P_4$

836

10) $5 \cdot {}_6P_5$

3,600

Find the number of unique permutations of the letters in each word.

11) DESIGN

720

12) MATH

24

13) CHEESE

120

14) FURTHER

2,520

15) BALLISTICS

453,600

16) BILLIONAIRE

3,326,400

Critical thinking questions:

17) Write a word for which there are 30 unique permutations of the letters.

LEVEL

18) Simplify ${}_xP_2$ $x(x-1)$